VECTOR CALCULUS, LAPLACE TRANSFORM & NUMERICAL METHODS (MA221TA)

UNIT-II

VECTOR INTEGRATION

TUTORIAL SHEET - I

1. The value of the line integral $\int_C (y^2 dx + x^3 dy)$ where c is the boundary of the square $-1 \le x \le 1, -1 \le y \le 1$ is_____ Ans: 0
2. Let ϕ be a scalar potential function of a conservative field $\vec{F}$. Then the work done in moving a particle along a curve C from point A to B is _____ Ans: $\phi(B) - \phi(A)$
3. The flux of the field $\vec{F} = x^2 i + yj$ around and across the closed curve $\vec{r} = \cos(t)\, i + \sin(t)\, j, 0 \le t \le 2\pi$ is ____ Ans: π
4. The acceleration of a moving particle at time t is given by $\vec{a} = 9t^2 i - 24t^2 j + 4\sin t\, k$. If $\vec{r} = 2i + j$ and velocity $\vec{v} = -i - 3k$ at $t = 0$. Find $\vec{r}$.
   Ans: $\left(\frac{3t^4}{4} - t + 2\right)\hat{\imath} - (2t^4 - 1)\hat{\jmath} + (t - 4\sin t)\hat{k}$
5. Find the total work done by the force represented by $\vec{F} = 3xy\hat{\imath} - y\hat{\jmath} + 2zx\hat{k}$ in moving a particle round the circle $x^2 + y^2 = 4, x = 2\cos\theta, y = 2\sin\theta$ & $z = 0, 0 \le \theta \le 2\pi$.
6. Evaluate $\int_C y^2 dx - 2x^2 dy$ along the parabola y = x^2 from (0,0) to (2,4).
7. Evaluate the line integral $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$, where $C: square: x = \pm 1, \quad y = \pm 1 \quad Ans: 0$
8. If $\vec{F} = (4xy - 3x^2z^2)i + 2x^2 j - 2x^3 zk$, show that the integral $\int_C \vec{F}.d\vec{r}$ is independent of the path c.
9. Find $\int_C \vec{F}.d\vec{r}$ where c is the $x^2 + y^2 = 4$ in the $\vec{F} = (2xy + z^3)i + x^2 j + 3xz^2 k$. Ans: 0.

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VECTOR INTEGRATION

TUTORIAL SHEET – II

1. The value of the surface integral $\iint_S (yz\,dy\,dz + zx\,dz\,dx + xy\,dx\,dy)$ where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ _____ Ans:0
2. Let S be a closed orientable surface enclosing a unit volume. Then the magnitude of the surface integral $\int\int_S \vec{r}.\hat{n}\,ds$, where $r = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and $\hat{n}$ is the unit normal to the surface S, equals _____ Ans: 3
3. If $\vec{f} = ax\hat{\imath} + by\hat{\jmath} + cz\hat{k}, a, b, c$ constants then $\iint f.dS$ where S is the surface of a unit sphere, is_____ Ans: $\frac{4}{3}\pi(a+b+c)$
4. By using Green's theorem, the area of an ellipse $\frac{x^2}{25} + \frac{y^2}{289} = 1$ is _____ Ans: $85\,\pi$
5. Verify Green's theorem for $\int_C (e^{-x}\sin y)dx + (e^{-x}\cos y)dy$, where C is the rectangle, whose vertices are $(0,0), (\pi,0), \left(\pi,\frac{\pi}{2}\right)$ and $\left(0,\frac{\pi}{2}\right)$. Ans: $[2(e^{-\pi} - 1)]$
6. Using Green's theorem, evaluate $\oint_C (x^2 - \cosh y\,dx + (y + sinx)dy)$ where C is the boundary of the rectangle $0 \le x \le \pi,\ 0 \le y \le 1$. $Ans: \pi(\cosh 1 - 1)$
7. Using the Green's theorem ,find the area enclosed between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$
8. If S is the surface of the sphere $x^2 + y^2 + z^2 = d^2$ and $\vec{A} = ax\hat{\imath} + by\hat{\jmath} + cz\hat{k}$, evaluate $\iint_S \vec{A}\cdot\hat{n}\,ds$. $Ans: \frac{2\pi d^3}{3}(a+b+c)$
9. If $\vec{F} = 2y\hat{\imath} - 3\hat{\jmath} + x^2\hat{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$, show that $\iint_S \vec{F}\cdot\hat{n}\,ds = 132$.
10. Find the surface integral over the parallelepiped $x = 0, y = 0, x = 1, y = 2, z = 3$ when $\vec{A} = 2xy\hat{\imath} + yz^2\hat{\jmath} + xz\hat{k}$ $Ans: 33$.

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UNIT-II

VECTOR INTEGRATION

TUTORIAL SHEET - III

1. Using divergence theorem, evaluate $\iint_S \vec{r}.\hat{n}\, ds$ where S is the surface of the sphere$x^2 + y^2 + z^2 = 9$. $Ans: 108\pi$
2. Verify divergence theorem for $\vec{F} = 4xz\hat{\imath} - y^2\hat{\jmath} + yz\,\hat{k}$ taken over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.
3. Using divergence theorem, evaluate $\iint_S \vec{F} \cdot \hat{n}\, ds$ over the entire surface S of the region above xy plane bounded by the cone $x^2 + y^2 = z^2$ the plane $z = 4$ where $\vec{F} = 4xz\hat{\imath} - xyz^2\hat{\jmath} + 3z\,\hat{k}$ $Ans: 704\pi$
4. Verify divergence theorem, for $\vec{F} = 2x^2y\vec{\imath} - y^2\vec{\jmath} + 4xz^2\vec{k}$ taken over the region in the first octant bounded by $y^2 + z^2 = 9$ and $x = 2$ Ans:180
5. Verify Stokes's theorem where $\vec{A} = (2x - y)\hat{\imath} - yz^2\hat{\jmath} - y^2z\,\hat{k}$ and S: the upper half of the surface of the sphere $x^2 + y^2 + z^2 = 1$ $Ans: \pi$
6. Evaluate $\oint_C xy\, dx + xy^2\, dy$ by Stoke's theorem where C is the square in the xy plane with vertices $(1,0)$ $(-1,0)$ $(0,1)$ $(0,-1)$.
7. Evaluate $\oint_C 4z\, dx - 2x\, dy + 2x\, dz$ by Stoke's theorem where C is the ellipse $x^2 + y^2 = 1,\ z = y + 1$. $Ans: -4\pi$
8. Evaluate $\int_C \vec{F}.d\vec{r}$ using Stoke's theorem, where $\vec{F} = y\hat{\imath} + xz^3\hat{\jmath} - zy^3\,\hat{k}$, C is the circle $x^2 + y^2 = 4,\ z = 1.5$. Ans= $\frac{19}{2}\pi$